

UNIVERSITY OF CAMBRIDGE

What Does a (Lensed) Galaxy Look Like?

Reproducing and Validating Score-Based Posterior Reconstruction
in Strong Gravitational Lensing

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Abstract

Score-based diffusion models provide flexible learned priors for strong lensing source reconstruction, but their reproducibility and statistical reliability require independent evaluation. We reproduce the posterior reconstruction method of Adam et al. using an independently implemented pipeline trained on PROBES galaxy images, and extend the original analysis with multi-source residual tests, PQMass distribution comparisons, and MIRA posterior validation. The central results were reproduced: unconditional samples exhibited galaxy-like morphologies, posterior draws reconstructed the dominant source structure, and an out-of-distribution experiment reproduced the transition from likelihood-driven to prior-dominated inference as the observation noise increased. Across 16 simulated sources, most reconstructions achieved image-plane residual statistics close to the expected noise level, with a median source-level χ^2/N of 1.058. Residual energy varied across sources and was strongly associated with lensed-image flux within the analysed sample. PQMass found evidence against equality of the generated and selected PROBES distributions across the tested representations, while a fixed real-versus-real control did not detect a discrepancy. The MIRA score for the reconstructed posterior was compatible with its finite-sample matched reference at the available precision, although the assessment was limited by the small number of truth–observation pairs and their shared simulation settings. Overall, the pipeline reproduced the central method, while the additional diagnostics identified source-dependent residual performance and a mismatch between the generated prior and selected PROBES images beyond visual and image-plane agreement.

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1 Introduction

Strong gravitational lensing occurs when the gravitational field of a foreground mass produces highly distorted or multiple images of a more distant source. Galaxy-scale strong lenses provide information about the distribution of luminous and dark matter, cosmological distances, and distant sources that would otherwise be too faint or too small to study at comparable resolution [1]. Forecasts for wide-field surveys have predicted increases of up to three orders of magnitude in number of discoverable galaxy–galaxy lenses, strengthening the need for scalable and reliable analysis methods [2].

Source Reconstruction. Extracting this information requires disentangling the source’s intrinsic surface-brightness distribution from the distortion produced by the foreground lens. Source reconstruction therefore extends beyond simple image recovery: in a complete lens analysis, the source and lens potential are coupled, so source-model assumptions can propagate into lens-level inferences [3, 4]. This work isolates the source-reconstruction task by assuming a fixed, known lens model. Defining x as the source-plane image and y the observed image-plane data, the forward model is

$$y = Ax + \eta, \tag{1}$$

where A is the fixed source-to-image lensing operator and η denotes observational noise. The inferential target is $p(x \mid y)$, rather than a single best-fitting source image. The fixed lens assumption provides a controlled setting in which the source posterior sampler can be studied separately from uncertainty in the lens model.

Even with a fixed lens, this inverse problem remains inherently ill-posed. Magnification and multiple imaging constrain some source regions strongly, while other regions are only weakly constrained; noise and finite resolution remove further information. Pixelated source models offer greater morphological flexibility than low-dimensional parametric profiles, but they require regularisation or prior information. Semi-linear inversion provides an efficient reconstruction for a pixelated source under a fixed lens model [5], while Bayesian treatments show that the preferred regularisation can depend on the underlying source morphology [3]. The prior can therefore influence both the recovered morphology and the reported uncertainty precisely where the observation contains the least information.

Score-Based Diffusion Priors. Adam et al. proposed representing this prior with a score-based diffusion model trained on images of undistorted galaxies [6]. Score-based models learn the gradient of a continuous sequence of noise-corrupted data distributions, and use the estimated score to define a reverse-time stochastic differential equation [7]. In the lensing application, the learned prior score is combined with a likelihood score supplied by a differentiable forward model to generate approximate posterior source samples. Adam et al. report independent posterior samples, fits to the image-plane that almost reach the assumed noise level, and the expected transition from a likelihood-driven to a prior-dominated reconstruction for a source that is out of distribution [6]. These are treated here as published claims to be tested, rather than as assumptions about the reproduced pipeline.

The probabilistic formulation is important because a single reconstructed image can appear plausible while concealing substantial ambiguity. Posterior samples can distinguish structure that is consistently required by the observation from features that vary across plausible source configurations allowed by the prior and likelihood. However, visual agreement and image-plane residuals do not by themselves establish that the learned prior represents the galaxy population or that the posterior uncertainty is well calibrated. Reproducing the reported reconstruction behaviour establishes the central reproduction result. The additional diagnostics then assess the reliability of the learned prior and posterior beyond the validation performed in the original study.

Objectives of This Work. The primary objective of this report is to investigate the reproducibility of the score-based posterior source reconstruction method of Adam et al. using an independently implemented pipeline. The reproduced analysis includes preprocessing galaxy images from the PROBES

compilation [8], training a score-based prior, constructing a differentiable lensing forward model, drawing posterior source samples, evaluating image-plane residuals, and repeating the out-of-distribution noise experiment. The principal published claims are assessed across multiple sources rather than through a single selected visual example, and every outcome is treated as an informative result to be documented in full.

Building on this reproduction, the secondary objective is to extend the original validation. PQMass is used to test whether unconditional samples from the learned prior are statistically consistent with the real PROBES galaxy images [9]. MIRA is then used to assess the conditional posterior across simulated truth–observation pairs for which the true source is known [10]. These extensions address two distinct questions: whether the learned prior represents the empirical galaxy-image distribution, and whether the inferred posterior places probability mass appropriately around the truth.

The central question of the report is therefore: to what extent can the principal results of Adam et al. be independently reproduced, and what explains any discrepancies? A secondary question is whether distributional tests support the reliability of the learned prior and posterior beyond visual and residual-based checks. Section 2 introduces the strong lensing forward model, Bayesian source reconstruction, score-based diffusion models, and the validation methods used in this work. Section 3 describes the independent implementation and experimental setup. Section 4 presents the reconstruction and validation results. Section 5 discusses the reproducibility verdict, possible sources of discrepancy, and future work. Section 6 summarises the main findings and concludes.

2 Background

This section introduces the physical and probabilistic background required to evaluate the reproduction of Adam et al. It first describes strong lensing as an inverse problem, then develops the Bayesian and score-based formulation used for posterior source reconstruction. The final section introduces the distributional validation tools applied after the central reproduction analysis.

2.1 Strong Lensing as a Forward Model

A foreground mass deflects light from a background source, allowing multiple light paths to reach the observer and producing arcs or multiple images. Figure 1 summarises both the physical geometry and its computational interpretation. The source image x is mapped through the fixed lens operator A to the observed image y , while source reconstruction seeks the posterior distribution $p(x | y)$ [1].

Magnification is not uniform. A caustic is a curve in the source plane at which the magnification of an ideal point source formally diverges. A source close to or across a caustic can produce highly magnified arcs and multiple images. Those regions are strongly constrained by the data, whereas regions far from a caustic, or outside the part of the source mapped into the image, can remain weakly constrained. The reconstruction therefore has spatially varying uncertainty.

For a fixed lens model, lensing a pixelated source is a linear operation. Following the notation of Adam et al. [6], the simulated observation is

$$y = Ax + \eta, \quad \eta \sim \mathcal{N}(0, \sigma_N^2 I_m), \quad (2)$$

where $x \in \mathbb{R}^n$ is the source, $y \in \mathbb{R}^m$ is the observed image, and $A \in \mathbb{R}^{m \times n}$ is the lensing operator. The operator maps image-plane rays to source-plane coordinates, interpolates the source brightness, and applies the pixelisation used by the observation [11]. Gravitational lensing conserves surface brightness, although it changes the apparent area and total flux of an image.

The reproduced experiments use a singular isothermal ellipsoid with external shear. By fixing the lens model, we isolate the source-inference problem from uncertainties in the lens parameters, providing a controlled test of source reconstruction. In a full analysis, the two are mathematically entangled; an overly restrictive source model can mistakenly transfer intrinsic source structure into the inferred lens or obscure perturbations induced by dark matter substructure [3, 4]. Consequently, the

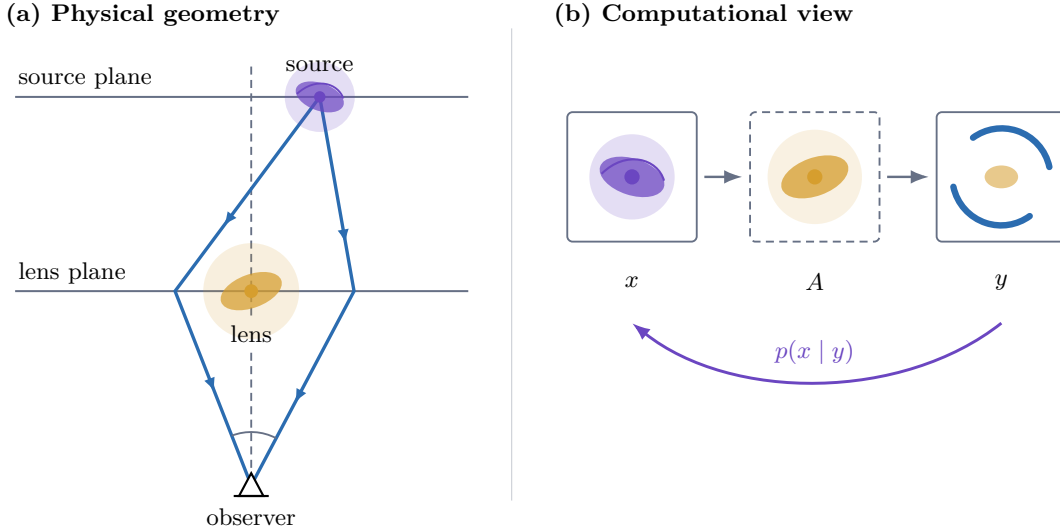


Figure 1: Physical and computational views of strong gravitational lensing. Panel (a) shows two deflected light paths connecting a background source to the observer through a foreground lens. Panel (b) represents the source to image mapping through the fixed lens operator A and the inverse target $p(x | y)$.

reliability of the source prior is critical, as a plausible image reconstruction alone does not guarantee an unbiased lens model.

2.2 Bayesian Source Reconstruction

The inverse problem is ill-posed because noise, finite resolution, and incomplete lensing coverage allow multiple sources to explain the same observation. Bayes' theorem combines the information in the observation with prior knowledge of plausible galaxies:

$$p(x | y) \propto p(y | x)p(x), \quad (3)$$

Under the Gaussian noise model, the likelihood and its gradient with respect to a clean source are

$$p(y | x) = \mathcal{N}(y | Ax, \sigma_N^2 I_m), \quad (4)$$

$$\nabla_x \log p(y | x) = \sigma_N^{-2} A^\top (y - Ax). \quad (5)$$

The transpose operator therefore maps an image-plane residual back into the source plane. Classical pixelated reconstructions control the unconstrained degrees of freedom with an explicit regularisation penalty [5, 3]. Adam et al. instead use a learned probability distribution over galaxy images as the prior [6].

The inferential target is the full posterior. A posterior mean can blur features whose locations or shapes vary across plausible sources. Individual samples retain those alternatives, while their variation shows where the observation is informative and where the result is mainly determined by the prior.

Data consistency is summarised in this project by

$$\frac{\chi^2}{N} = \frac{1}{N} \frac{\|y - Ax\|_2^2}{\sigma_N^2}, \quad (6)$$

where $N = m$ is the number of image pixels included in the calculation. For the true source under independent Gaussian noise, its expectation is one. Without a well-defined effective number of degrees of freedom, we denote this statistic as *chi-squared per pixel* or *normalised image-plane residual energy*

to distinguish it from a true reduced chi-squared. While a value near unity indicates consistency with the assumed noise level, it does not inherently guarantee the accuracy of the uncertainty in the source space. Related learned approaches include recurrent inference machines, continuous neural fields, and diffusion restoration models [12, 13, 14, 15].

2.3 Score-Based Diffusion Models as Galaxy Priors

The score at diffusion time t is

$$s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t). \quad (7)$$

It points in the local direction of increasing probability density. Song et al. formulate the forward process as the Itô SDE

$$dx = f(x, t) dt + g(t) dw, \quad (8)$$

where f is the drift, g is the diffusion coefficient, and w is a standard Wiener process. Starting from data $x_0 \sim p_0$, the process gradually approaches a tractable noise distribution. Sampling reverses this evolution:

$$dx = [f(x, t) - g^2(t) \nabla_x \log p_t(x)] dt + g(t) d\bar{w}, \quad (9)$$

where time runs from $t = 1$ to $t = 0$. The reverse drift depends on the unknown density only through its score, so the density and its normalising constant never need to be evaluated directly [16, 7].

The reproduced model uses the variance exploding case, for which $f(x, t) = 0$. Its noise schedule is

$$\sigma(t) = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^t, \quad x_t = x_0 + \tau(t)z, \quad (10)$$

$$z \sim \mathcal{N}(0, I), \quad \tau^2(t) = \sigma^2(t) - \sigma^2(0). \quad (11)$$

The network is trained by denoising score matching. In general form, the objective is

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, z} \left[\lambda(t) \left\| s_\theta(x_t, t) + \frac{x_t - x_0}{\tau^2(t)} \right\|_2^2 \right], \quad (12)$$

where $\lambda(t)$ weights different noise levels. The Gaussian perturbation has the analytic score $-(x_t - x_0)/\tau^2(t)$, and denoising score matching has the required marginal score as its population optimum [17, 18, 19]. For this schedule, $g^2(t) = d\sigma^2(t)/dt$. Under the exact VE SDE convention, the accumulated perturbation variance is $\tau^2(t) = \sigma^2(t) - \sigma_{\min}^2$. The implementation follows the `score_models` convention and uses $\sigma^2(t)$ directly in the convolved likelihood. The two differ by $\sigma_{\min}^2 = 10^{-8}$ for the trained model.

Replacing the unknown score in the reverse SDE by s_θ gives unconditional samples from the learned prior. The implementation uses an NCSN++ network and PROBES galaxy images [7, 8]. This prior represents the distribution of the preprocessed training data and may not generalise to the broader population of physical galaxies, requiring its fidelity to be assessed empirically.

2.4 Posterior Sampling with a Learned Prior

At diffusion time t , Bayes' rule decomposes the posterior score as

$$\nabla_{x_t} \log p_t(x_t | y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(y | x_t). \quad (13)$$

The learned network supplies the first term. The second is not the clean source likelihood given above, since x_t contains diffusion noise. Adam et al. approximate the intermediate likelihood by convolving the observation noise with that diffusion noise:

$$p_t(y | x_t) \approx \mathcal{N}(y | Ax_t, \Sigma_t), \quad \Sigma_t = \sigma_N^2 I_m + \tau^2(t) AA^\top. \quad (14)$$

The resulting likelihood score is

$$\nabla_{x_t} \log p_t(y | x_t) = A^\top \Sigma_t^{-1} (y - Ax_t). \quad (15)$$

Direct inversion of Σ_t at every step is expensive. The reproduced sampler additionally assumes $AA^\top \approx I_m$, giving the scalar covariance $(\sigma_N^2 + \tau^2(t))I_m$ [6]. Surface brightness conservation does not make this identity exact. Interpolation introduces correlations between neighbouring image pixels, while pooling rescales the rows of A . Both operations may alter the effective data covariance, making empirical validation of this approximation necessary.

We discretised the posterior reverse SDE using the standard Euler–Maruyama method. Let $h_k = t_k - t_{k-1}$ denote the magnitude of a backward time step. Defining $s_{\text{post}} = s_\theta + \nabla_x \log p_t(y | x)$, one update is

$$x_{k-1} = x_k + g^2(t_k) s_{\text{post}}(x_k, t_k) h_k + g(t_k) \sqrt{h_k} z_k, \quad z_k \sim \mathcal{N}(0, I). \quad (16)$$

The drift combines the prior and likelihood information, while the Gaussian increment represents reverse diffusion noise. Independent initial noise and increments produce separate approximate posterior samples. The reverse-time update follows the convention of Song et al. and Adam et al. [7, 6]. Using a finite number of Euler–Maruyama steps introduces discretisation error. The fidelity of the resulting posterior samples is limited by several distinct approximations, summarised in Table 1.

Approximation	Possible consequence	Relevant diagnostic
Learned score replaces the true prior score	Missing or distorted galaxy morphology	Prior samples and PQMass
Convolved intermediate likelihood [6]	Incorrect balance between prior and observation	OOD noise sweep and residuals
$AA^\top \approx I_m$	Incorrect likelihood scale or correlations	Direct inspection of $AA^\top$ and pooling tests
Finite Euler–Maruyama steps	Numerical bias in posterior samples	Step count sensitivity

Table 1: Diagnostic separation of the main approximations in the sampler. Matching a single reconstructed image does not establish that each component behaves as reported.

2.5 Distributional Validation Extensions

Image-plane residuals test whether reconstructed sources explain the observation, but they do not establish that the learned prior matches the galaxy distribution or that the candidate posterior has the correct conditional distribution. Generative-model evaluation must account for both sample fidelity and distributional diversity, which may not be captured by individual visual examples [20]. PQMass and MIRA address these questions as extensions to the central reproduction analysis.

2.5.1 PQMass for Prior Validation

Sample-based two-sample tests compare distributions without requiring tractable probability densities [21]. PQMass is a likelihood-free two-sample test [9]. Given independent samples $\{u_i\}_{i=1}^m \sim p$ and $\{v_i\}_{i=1}^n \sim q$, independent reference points partition the representation space into n_R Voronoi regions $\{R_j\}_{j=1}^{n_R}$. Define the region counts

$$k_{uj} = \sum_{i=1}^m \mathbb{1}(u_i \in R_j), \quad k_{vj} = \sum_{i=1}^n \mathbb{1}(v_i \in R_j). \quad (17)$$

Under the null hypothesis $p = q$, the pooled probability assigned to region R_j is estimated by

$$\hat{p}_j = \frac{k_{uj} + k_{vj}}{m + n}, \quad \hat{N}_j^{(1)} = m\hat{p}_j, \quad \hat{N}_j^{(2)} = n\hat{p}_j. \quad (18)$$

Following the Pearson test used in the original method, the PQMass statistic is

$$\chi_{\text{PQM}}^2 = \sum_{j=1}^{n_R} \left[\frac{\left(k_{uj} - \hat{N}_j^{(1)}\right)^2}{\hat{N}_j^{(1)}} + \frac{\left(k_{vj} - \hat{N}_j^{(2)}\right)^2}{\hat{N}_j^{(2)}} \right]. \quad (19)$$

Under the null hypothesis and for adequate expected counts, the observed statistic χ_{PQM}^2 is approximately distributed as a chi-squared random variable with $n_R - 1$ degrees of freedom, giving

$$p_{\text{PQM}} = \int_{\chi_{\text{PQM}}^2}^{\infty} f_{\chi_{n_R-1}^2}(z) dz. \quad (20)$$

A small p_{PQM} provides evidence against equality of the quantised distributions. A large value indicates only that no difference was detected at the available sample size.

2.5.2 MIRA for Posterior Validation

Simulation-based posterior validation uses known truths and simulated observations to test whether posterior samples have the expected coverage behaviour [22, 23]. MIRA evaluates a candidate conditional distribution using joint samples from the true data-generating process [10].

For a truth and observation pair (x^*, y^*) , let $\{x_j\}_{j=1}^N \sim q(x | y^*)$ be candidate posterior samples. Given an independent reference sample $x_r \sim q(x | y^*)$ and a random centre $c \sim p(c)$, MIRA constructs the region

$$\mathcal{R} = \{x \in \mathcal{X} : d(x, c) \leq d(x_r, c)\}. \quad (21)$$

The candidate count and truth indicator are

$$n = \sum_{j=1}^N \mathbb{1}(x_j \in \mathcal{R}), \quad k = \mathbb{1}(x^* \in \mathcal{R}). \quad (22)$$

Under the null hypothesis $q(x | y) = p(x | y)$, the MIRA statistic follows Laplace's rule of succession:

$$p(k | n) = \frac{n+1}{N+2} \mathbb{1}(k=1) + \frac{N-n+1}{N+2} \mathbb{1}(k=0). \quad (23)$$

Let Z collect the randomness in the truth and observation pair, candidate samples, centre, and region. The population MIRA score is

$$\mu_{\text{MIRA}}(q) = \mathbb{E}_{p(Z)} [\mathbb{E}_{p(k,n|Z)} [p(k | n)]] . \quad (24)$$

Using T independent truth and observation pairs and L_R regions per pair, it is estimated by

$$\hat{\mu}_{\text{MIRA}} = \frac{1}{TL_R} \sum_{t=1}^T \sum_{\ell=1}^{L_R} p(k_{t\ell} | n_{t\ell}). \quad (25)$$

For a correctly specified candidate distribution,

$$p(k | n) \xrightarrow[N \rightarrow \infty]{d} \text{Beta}(2, 1), \quad \mathbb{E}[p(k | n)] \rightarrow \frac{2}{3}, \quad \text{Var}[p(k | n)] \rightarrow \frac{1}{18}. \quad (26)$$

The matched distribution reference is therefore $2/3$, with theoretical standard uncertainty $\sqrt{1/(18T)}$ for T independent truth and observation pairs [10]. Drawing more regions reduces Monte Carlo error but does not increase the number of independent pairs. Scores below $2/3$ are consistent with overconfidence or bias, while scores above $2/3$ indicate underconfidence under the variance interpretation of Sharief et al. Crucially, agreement with this reference value is necessary but not sufficient to establish

full posterior calibration. Posterior draws and spatial regions within the same truth–observation pair are correlated, so calibration must be assessed across multiple independent pairs and scientifically relevant test quantities [24, 25].

PQMass and MIRA have complementary roles. PQMass tests the learned marginal prior $p(x)$, whereas MIRA tests the candidate conditional posterior $q(x | y)$. Both are extensions applied after the reproduction of Adam et al.; neither changes the main reproducibility question or replaces direct reconstruction and residual evidence.

3 Method and Implementation

We implemented the score-based source-reconstruction pipeline of Adam et al. [6]. The reproduction comprised data preparation, training the galaxy prior, simulating lensed observations, posterior sampling, and an out-of-distribution noise experiment. PQMass and MIRA were then applied as extensions to test the learned prior and posterior more directly. Their mathematical motivation is given in Section 2; this section specifies how each experiment was carried out.

3.1 Data Preparation

The prior was trained on g -band galaxy images from the PROBES survey [8]. Raw FITS files were retrieved with the public `astroddpm` download script associated with Smith et al. [26].¹ All filtering, band selection, cropping, and normalisation were implemented independently in this project. Files smaller than 5000 bytes were rejected, as were images containing non-finite values or more than 30 percent zero-valued pixels. Each accepted image was centre cropped to 256×256 pixels. For an original pixel flux u , the stored value was

$$\bar{u} = 2 \frac{\text{clip}(u, 0, 5.5)}{5.5} - 1. \quad (27)$$

This transformation follows the flux range used by Smith et al. and maps every input to $[-1, 1]$ [26]. A fixed random selection of the resulting preprocessed images is shown in Figure 2.

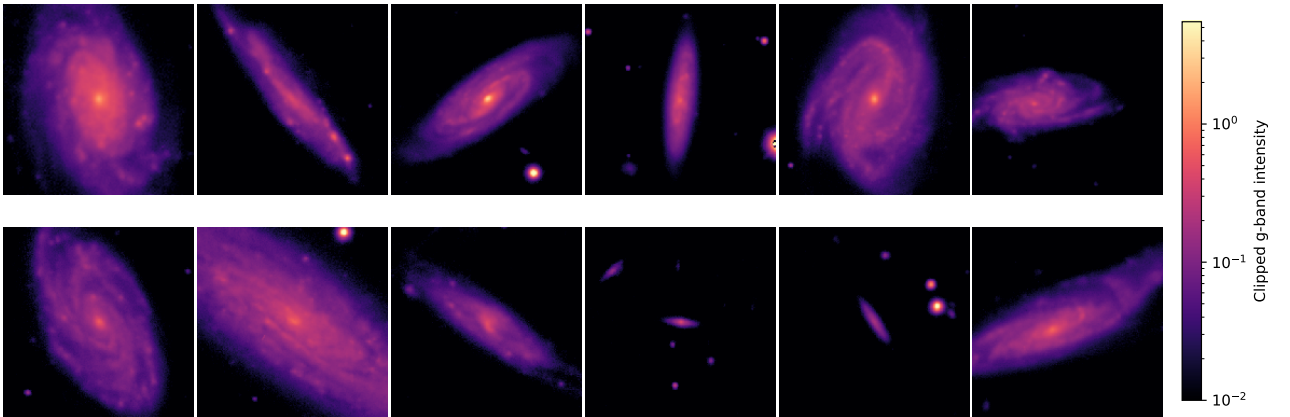


Figure 2: Random selection of 12 preprocessed PROBES g -band galaxies from the training catalogue, shown as 256×256 centre crops with a common logarithmic intensity scale.

The full processed catalogue was used to train the prior. The PROBES galaxies later selected as simulated truths represent specific evaluation instances rather than a conventional test set, which must be taken into account when interpreting the resulting reconstructions.

¹https://github.com/Smith42/astroddpm/blob/master/data/probes/get_probes.sh, accessed 1 July 2026.

3.2 Score-Based Prior Training

Our implementation utilises the NCSN++ architecture, variance exploding SDE interface, and denoising score matching loss from `score_models` version 0.5.11 [7, 6].² Pipelines for distributed data loading, optimisation, exponential moving averages, and checkpoint resumption were developed independently. Since Adam et al. do not report enough architectural detail to establish an exact match, we configured our NCSN++ model with 128 base channels and channel multipliers [1, 1, 2, 2, 2, 2, 2].

Training used all processed images, distributed over four A100 GPUs. The effective batch size and optimisation budget closely match the values reported by Adam et al. We employed the exponential moving average weights stored in the final checkpoint for all subsequent sampling experiments. Table 2 summarises the training configuration and the main reproduction comparison.

Setting	Adam et al.	This work
Image size	256×256	256×256
Network	NCSN++	NCSN++
n_f	not reported	128
Channel multipliers	not reported	[1, 1, 2, 2, 2, 2, 2]
$\sigma_{\min}$	10^{-4}	10^{-4}
$\sigma_{\max}$	263.4	263.4
Optimiser	Adam	Adam
Learning rate	not reported	10^{-4}
EMA decay	not reported	0.9999
Effective batch size	16	16 (4 per GPU $\times$ 4 GPUs)
Training length	approximately 350,000	2700 epochs, $\sim$ 351k steps
Warmup	not reported	5000 steps
Gradient clipping	not reported	norm 1.0
Hardware and wall time	4 V100 GPUs, $\sim$ 70 h	4 A100 GPUs, $\sim$ 24 h

Table 2: Prior-training configuration compared with the settings reported by Adam et al.

3.3 Lensing Simulations

Mock observations were generated with the forward model defined in Section 2.1. The differentiable simulator was implemented with `caustics` [27]. The deflector comprised a singular isothermal ellipsoid [28] and external shear, with fixed parameters

$$z_l = 0.5, \quad z_s = 1.5, \quad q = 0.7, \quad \phi = \frac{\pi}{6}, \quad R_{\text{Ein}} = 1.2 \text{ arcsec}, \quad \gamma_1 = 0.03, \quad \gamma_2 = 0.04. \quad (28)$$

The source grid contained 256×256 pixels at 0.028 arcsec per pixel. Lensed images were first rendered on a 256×256 grid at 0.04 arcsec per pixel, then average pooled by a factor of two to form 128×128 observations. Independent Gaussian noise was added after pooling. The main reconstruction experiments used $\sigma_N = 10^{-2}$, matching Adam et al. [6]. The same lens configuration was used for simulation and inference, isolating source reconstruction from lens-model error.

3.4 Posterior Sampling

For each reconstruction, a preprocessed PROBES image was treated as the true source x^* . The fixed forward model produced a clean lensed image, which was pooled and perturbed once to obtain the stored observation y . Reusing this stored observation ensured that resumed sampling jobs remained conditioned on the same data.

²https://github.com/AlexandreAdam/score_models, accessed 11 March 2026.

Posterior samples were generated with the convolved-likelihood reverse-SDE approximation of Adam et al. [6]. At diffusion time t , the implementation used the scalar likelihood variance

$$\sigma_N^2 + \sigma^2(t), \quad (29)$$

and differentiated the likelihood through the `caustics` simulator with automatic differentiation. This is the implemented form of the $AA^\top \approx I_m$ approximation discussed in Section 2.4.

The standard posterior configuration used 8000 Euler–Maruyama steps and produced 160 draws in five chunks of 32. The observation was saved before sampling, and chunk c was generated with seed $21 + c$ and written to disk on completion. On resumption, existing chunks were skipped. The script checked the source, noise level, and base seed; consistency of the checkpoint, solver, lens configuration, and code version was maintained manually.

All source runs used the same base seed of 21. The simulated observations therefore share one noise realisation, and posterior sampling uses the same sequence of chunk seeds for every source. Comparisons across sources are consequently conditional on this fixed random realisation, and the resulting truth-and-observation pairs should not be interpreted as independent noise replicates.

For each posterior draw $x^{(s)}$, image-plane consistency was measured by

$$q_s \equiv \frac{\chi_s^2}{N} = \frac{\|y - Ax^{(s)}\|_2^2}{N\sigma_N^2}, \quad N = 128^2, \quad (30)$$

where A includes the same lensing and average-pooling operations used to generate the observation. The statistic was also evaluated for the injected true source and the posterior-mean source. For each source, the posterior-draw distribution was summarised by its median and 16th–84th percentile interval. The complete distribution was retained, and no draws were selected on the basis of their residuals.

To examine whether image-plane agreement depended on source brightness, the statistic was compared with the clean lensed-image flux. Reversing the training normalisation gives

$$F_{\text{img}} = \sum_{j=1}^N \frac{5.5}{2} (y_{\text{clean},j} + 1). \quad (31)$$

The association between $\log F_{\text{img}}$ and the median posterior-draw χ^2/N was assessed with Spearman’s rank correlation. Its conditional uncertainty over the finite source set was summarised by the 2.5th and 97.5th percentiles from 2000 bootstrap resamples of the source identities.

3.5 Out-of-Distribution Noise Experiment

The OOD source was independently generated using OpenAI image generation tool in ChatGPT and the prompt “A galaxy in the shape of the number 7 on a dark background”, matching the prompt reported by Adam et al. [6]. Because the resulting image is not identical to the source used by Adam et al., this experiment is a prompt-matched qualitative reproduction. The same fixed lens model was applied at noise levels

$$\sigma_N \in \{0.001, 0.1, 0.8, 1.0, 5.0\}. \quad (32)$$

One posterior draw was generated at each noise level using 8000 reverse-SDE steps. The base seed was 21, incremented once for each noise level. The resulting sequence was used to examine how the reconstructed morphology changed as the observation became less informative.

3.6 Distributional Validation Extensions

PQMass and MIRA were introduced after the main reconstruction pipeline had been reproduced. Source-to-source variation in the image-plane residuals could not establish whether the observed behaviour arose from mismatch in the learned prior or from the approximate posterior sampler. The

two tests therefore serve as diagnostic extensions: PQMass tests unconditional samples from the learned prior, whereas MIRA evaluates posterior calibration across repeated truth-and-observation pairs [9, 10].

3.6.1 PQMass Design

For the primary comparison, we utilised 1024 unconditional prior samples—generated from the saved checkpoint using 2000 Euler–Maruyama steps—alongside 1024 real PROBES images selected under a base seed of 21. The PQMass statistic was initially computed on the flattened pixel values [9]. As a robustness check, we also evaluated a 50-component PCA representation fitted to the combined real and generated datasets, serving as a complementary sensitivity analysis rather than a substitute for the pixel space baseline.

Each evaluation utilised the public PQMass implementation configured with 100 reference regions, 200 re-tessellations, z-score normalisation, and a Gaussian reference fraction of 0.05. A second comparison, performed between two disjoint subsets of real PROBES images, served as a negative control. The real-vs-real control used the same settings and helped distinguish generated-vs-real mismatch from gross artefacts of the test configuration.

3.6.2 MIRA Design

Posterior inference was initially run for 30 sources. Following loss of access to those outputs, the final validation dataset was constructed by rerunning 16 sources, each with $S = 160$ retained draws, as reported in Section 4.5.2. All truths and posterior draws were projected into a common 50-component PCA space, which retained 99.61 percent of their combined variance. The PCA basis was fitted once to the combined stack and then held fixed across the compared posterior sets. Feature min–max normalisation was enabled in the public implementation, and its finite- S normalised score was retained for reporting.

The analysis uses seed 21, 200 internal Monte Carlo replications, and 200 bootstrap resamples over truths. For $S = 160$, the finite-sample matched reference on the package-normalised scale is $(2S + 1)/(3S) = 0.66875$, close to the asymptotic value of $2/3$. A synthetic Gaussian experiment first checks the matched reference and the sensitivity of the implementation to covariance changes. For two directional checks on the reconstructed galaxies, deviations from each source-specific posterior mean are rescaled by $1/\sqrt{3}$ and $\sqrt{2}$. The resulting covariances are one third and twice that of the original posterior, matching the relative covariance discrepancies used for the overconfident and underconfident cases of Sharief et al. [10]. These modified posteriors test sensitivity and are not calibration thresholds.

Because the compiled simulation pipelines shared a common noise realisation and aligned posterior sampling streams, MIRA is treated as a calibration diagnostic restricted to this fixed collection of simulated observations. Its bootstrap uncertainty is conditional on the saved simulations and does not represent the variation that would arise from independently regenerating every observation and posterior sample set.

3.7 Reproducibility

Preprocessing, prior training, lens simulation, and posterior sampling were organised in separate scripts. The `score_models` package supplied the NCSN++ architecture, variance-exploding SDE interface, and denoising score-matching loss described above [7, 6]. `astroddpm` was used only to retrieve the raw PROBES files; `caustics`, PQMass, and MIRA supplied the specialist simulation or validation routines. Training checkpoints preserve the model configuration and optimisation state, while posterior output files preserve the source identity, observation, noise level, solver step count, and samples. Exact commands and output locations are recorded in the repository README.

4 Results

4.1 Learned Prior Samples

Unconditional samples were generated from the saved EMA checkpoint without conditioning on a lensing observation. Figure 3 compares fixed random selections of these draws and real preprocessed PROBES galaxies [8] using the same inverse normalisation and display stretch. The generated samples reproduce centrally concentrated and elongated morphologies across a range of apparent sizes, orientations, and normalised brightnesses. Compared with the PROBES examples, they are generally smoother and contain less resolved small-scale structure, while occasional compact off-source features occur in both sets.

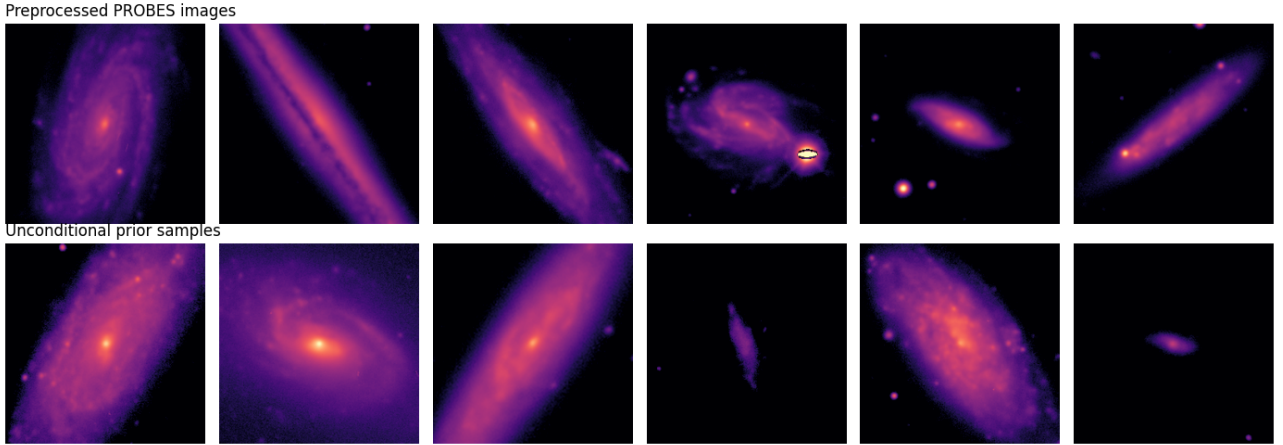


Figure 3: Examples from the preprocessed PROBES catalogue (top) and unconditional samples from the reproduced score-based prior (bottom). The two rows were sampled independently, so columns do not represent matched pairs. Images are inverse-normalised for display; quantitative comparisons are performed in the model input space.

A diagnostic showed broadly overlapping distributions of summed normalised pixel values for the real and generated samples. Their maximum pixel intensities were also similar, although both distributions contained a large concentration at upper clipping bound. These low-dimensional comparisons and the visual examples do not establish distributional agreement, which is assessed separately with PQMass.

4.2 Posterior Reconstruction

Following the layout of Figure 1 in Adam et al. [6], Figure 4 presents the reconstruction of Mathewson96 ESO297-G29 at $\sigma_N = 0.01$. The posterior draws recover the source orientation, central concentration, and main spiral structure, while differing in smaller-scale features. The posterior mean retains the overall extent and morphology but smooths the spiral arms. After lensing, the reconstructed sources reproduce the main Einstein-ring structure and the residuals are predominantly noise-like, with weak coherent structure remaining near the brightest regions. Results across all evaluated sources are quantified in Figure 5.

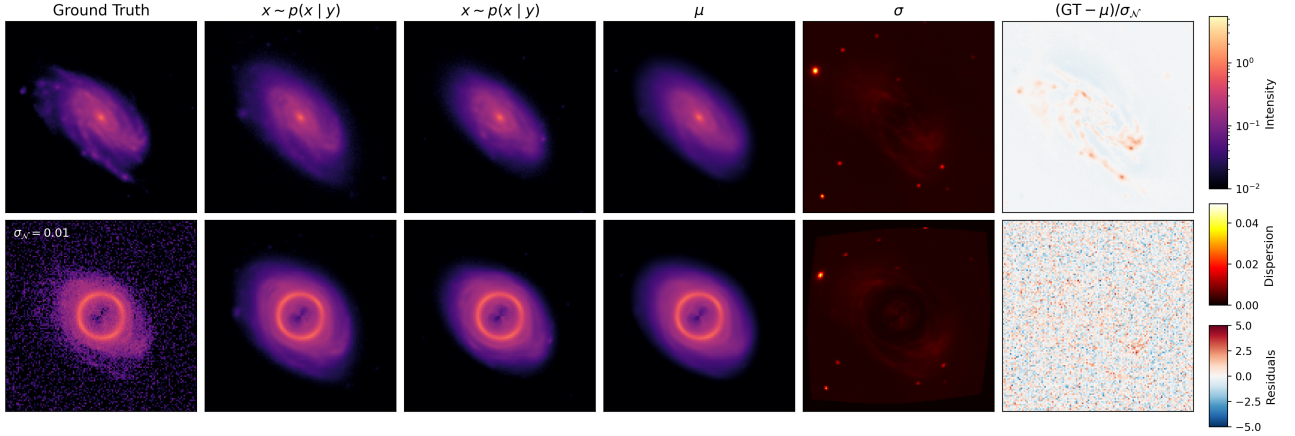


Figure 4: Source and observation reconstruction for Mathewson96 ESO297-G29 using the saved EMA checkpoint and 8000 Euler–Maruyama steps at $\sigma_N = 0.01$. The columns show the truth and observation, two posterior draws, the mean and standard deviation of 160 draws, and residuals in the source and image planes. Following Adam et al., the source-plane residual is displayed as $(GT - \mu)/\sigma_N$ for illustration only because the expected source reconstruction accuracy varies with position.

4.3 Image-Plane Residual Consistency Across Sources

Adam et al. report reconstructions that agree with the observation almost to the noise level [6]. This claim was tested across the 16 completed source runs available for analysis. For each source, χ^2/N was evaluated for the injected truth, the posterior-mean source, and all 160 posterior draws. Figure 5 shows the resulting source-to-source variation and its association with clean image-plane flux.

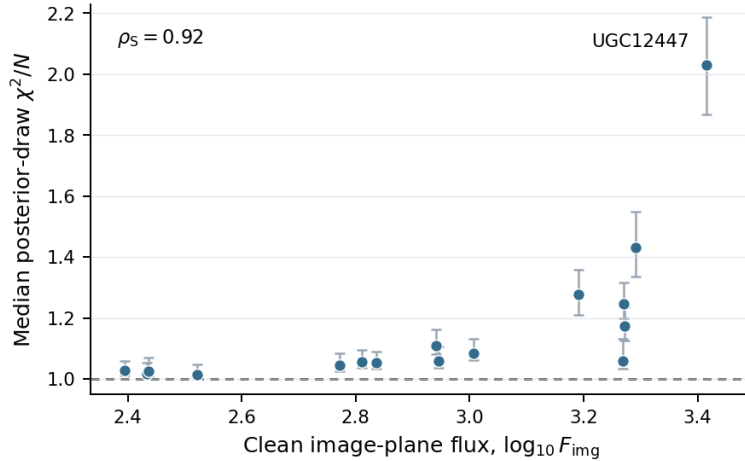


Figure 5: Image-plane residual energy as a function of clean lensed-image flux for the 16 completed source runs available for analysis. All sources were reconstructed at $\sigma_N = 0.01$ using 8000 Euler–Maruyama steps and 160 posterior draws. Points show the median χ^2/N across posterior draws, intervals show the 16th–84th percentiles, and the dashed line marks the noise reference $\chi^2/N = 1$.

Across the 16 sources for which inference was repeated after the original CSD3 outputs became inaccessible, the median of the 16 source-level posterior-draw medians is 1.058. Ten sources have a median no greater than 1.1, and 12 are below 1.2, showing that agreement close to the noise level is common but not uniform. UGC12447 is the clearest outlier: the statistic for its posterior mean is 1.891, while the median across its posterior draws is 2.030, with a 16th–84th percentile interval of [1.868, 2.186].

Within this set, the residual statistic shows a strong rank correlation with clean lensed image flux (Spearman’s $\rho = 0.924$; 95 percent bootstrap interval over sources: $[0.736, 0.982]$). Excluding UGC12447 gives $\rho = 0.907$, while estimates obtained by removing each source in turn range from 0.907 to 0.939. The association is therefore not produced by UGC12447 alone. Taken together, these checks support a consistent association between residual energy and flux across the analysed sources. Since the 16 sources available for reanalysis were determined by computational availability rather than a predefined sampling scheme, the strength of this relation should not yet be generalised to the wider PROBES population.

The true-source statistic is 0.989978 in every run because the same standardised noise realisation was reused across sources. The posterior-mean statistic is below the posterior-draw median for every source, consistent with projected posterior spread contributing to the per-draw residual in addition to systematic image-plane error. The multi-source result supports the published claim in Adam et al. for most tested sources and also shows that its accuracy is source dependent.

4.4 Out-of-Distribution Noise Sweep

Figure 6 shows one posterior draw at five noise levels for the independently generated, prompt-matched digit-7 source. At $\sigma_N = 0.001$ and 0.1, the digit remains recognisable in the reconstruction. At $\sigma_N = 0.8$ and 1.0, its continuous structure weakens and gives way to increasingly diffuse, galaxy-like features. At $\sigma_N = 5$, the digit is no longer recovered and the sample is dominated by the learned galaxy prior.

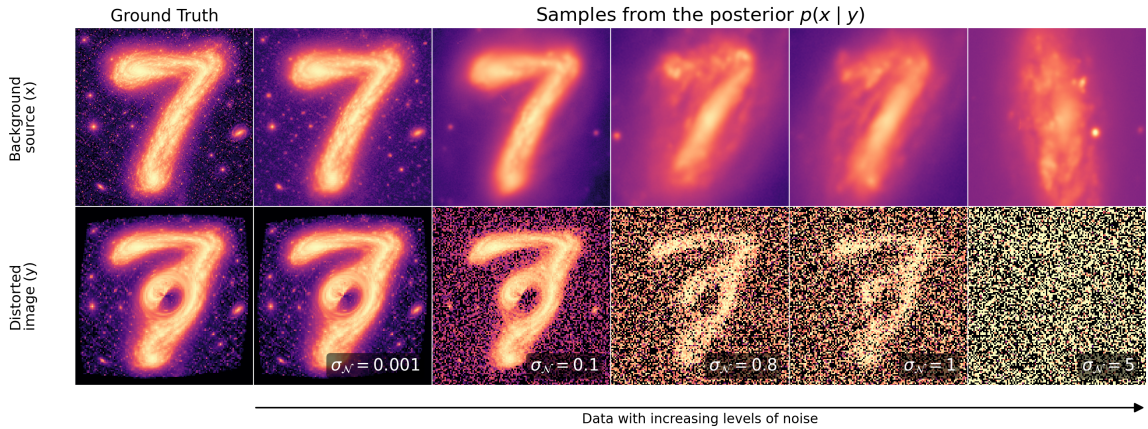


Figure 6: Out-of-distribution reconstruction across increasing observation noise. The leftmost column shows the digit-7 source and its noiseless lensed image. The source was generated in ChatGPT using OpenAI’s DALL-E image-generation system with the prompt used by Adam et al. The remaining columns show one posterior draw (top) and the corresponding observation (bottom) at different σ_N . Each draw used 8000 Euler–Maruyama steps, with seeds 21–25 assigned in order of increasing noise.

The sequence qualitatively reproduces the transition reported by Adam et al. [6]: an informative likelihood preserves the out-of-distribution structure, whereas the learned prior controls the reconstruction as the observation becomes less informative.

4.5 Distributional Validation Extensions

4.5.1 Prior Validation with PQMass

PQMass compared 1024 prior samples generated with 2000 reverse-SDE steps against 1024 real PROBES images in pixel space [9]. Each of 200 re-tessellations used 100 reference regions, giving 99 degrees of freedom. Under the null hypothesis, $\chi^2_{\text{PQM}}/99$ should be centred near one and the p-

values should be approximately uniform. Figure 7 compares the primary pixel-space result, its PCA sensitivity check, and the real-versus-real control.

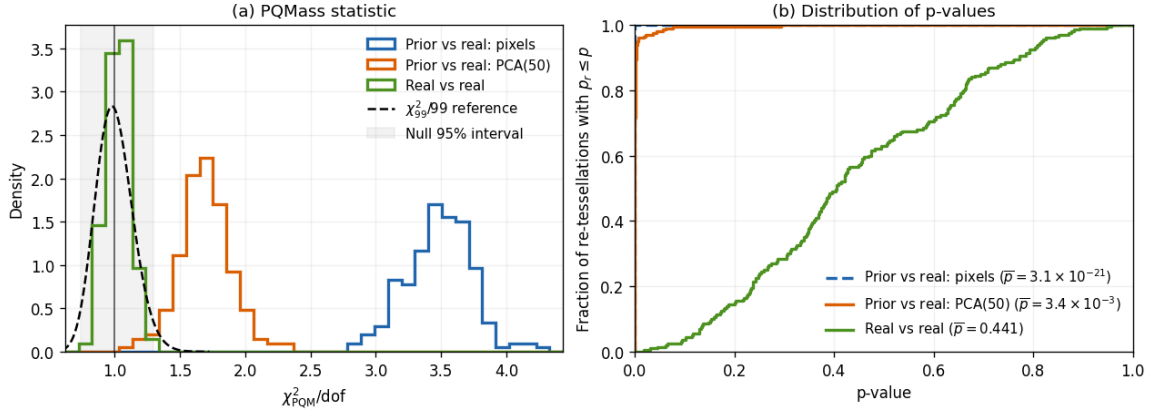


Figure 7: PQMass results across 200 re-tessellations of the fixed image sets. Panel (a) shows the distributions of $\chi^2_{\text{PQM}}/99$ for the pixel-space comparison, PCA(50) cross-check, and real-versus-real control. The dashed curve and shaded region show the corresponding null distribution and its central 95 percent interval. Panel (b) shows the empirical cumulative distributions of the resulting p -values. The re-tessellations vary the reference regions but reuse the same image sets; sample counts and numerical summaries are given in Table 3.

In pixel space, the central 68 percent of $\chi^2_{\text{PQM}}/99$ lies between 3.212 and 3.718, and even the largest p -value is only 4.92×10^{-19} . The discrepancy is stable to the tested reference-region tessellations.

The real-versus-real control used two disjoint halves of the selected real set, giving 512 images per group. Its mean $\chi^2_{\text{PQM}}/99$ is 1.021, with a 16th–84th percentile interval of $[0.930, 1.108]$, and its mean p -value is 0.441. For this fixed split, the same test configuration did not detect a discrepancy between the two real subsets.

Comparison	Samples per set	Mean $\chi^2_{\text{PQM}}/99$	Mean p -value
Real vs. real (disjoint control)	512	1.021	0.441
Generated vs. real (pixels)	1024	3.481	3.07×10^{-21}
Generated vs. real (PCA(50))	1024	1.695	3.38×10^{-3}

Table 3: Summary of the PQMass comparisons. Values are averaged over 200 re-tessellations of the same fixed sample sets. The real-versus-real control uses two disjoint subsets of the selected PROBES images.

Projection into a common 50-component PCA space retains 82.29 percent of the combined sample variance. The generated-versus-real discrepancy becomes smaller but remains: the mean $\chi^2_{\text{PQM}}/99$ is 1.695, with a 16th–84th percentile interval of $[1.522, 1.886]$, while the p -value mean and median are 3.38×10^{-3} and 2.25×10^{-5} , respectively. Although some tessellations yield substantially larger p -values, reaching 0.293, the low mean and median show that the discrepancy persists overall.

Both representations provide evidence against equality of the generated and selected PROBES distributions, with a substantially larger discrepancy in pixel space. The 200 values arise from re-tessellations of fixed sample sets and therefore do not represent independent generated-versus-real experiments. Visual plausibility at the level of individual samples is thus insufficient to establish that the learned prior reproduces the empirical image distribution.

4.5.2 Posterior Validation with MIRA

Posterior inference was initially completed for 30 PROBES sources on CSD3, but the outputs became inaccessible following a cooling-related outage. Sixteen sources were subsequently re-run, yielding a

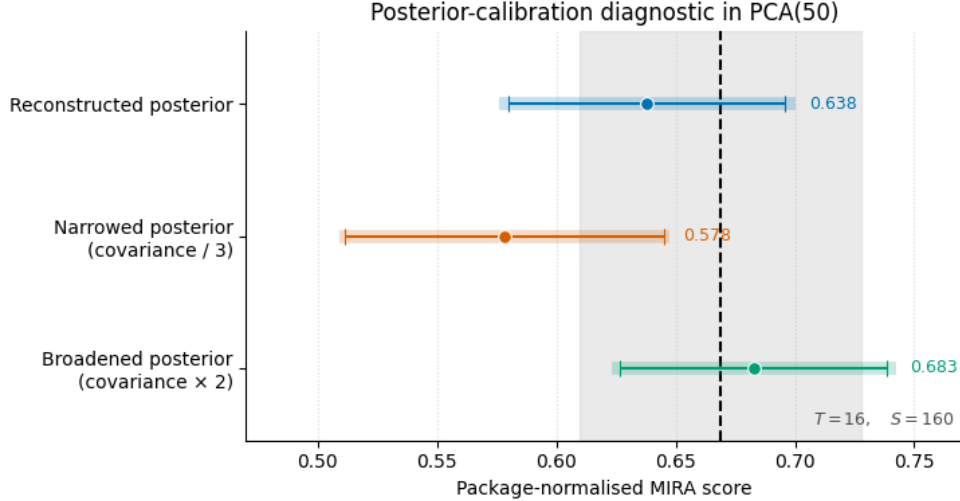


Figure 8: Comparison of package-normalised MIRA scores in the common PCA(50) representation. Points show the estimated scores, thin bars the Monte Carlo standard deviations, and broad translucent bars the bootstrap standard deviations. The dashed line and grey band indicate the finite-sample matched reference and its nominal spread. Exact values and baseline definitions are reported in Table 4.

Posterior	MIRA score	Monte Carlo s.d.	Bootstrap s.d.
Finite- S matched reference	0.66875	—	—
Reconstructed posterior	0.6379	0.0578	0.0621
Narrowed (overconfident)	0.5782	0.0668	0.0691
Broadened (underconfident)	0.6827	0.0559	0.0595

Table 4: Package-normalised MIRA results for $T = 16$ truths with $S = 160$ posterior draws each. The uncertainty columns report standard deviations from 200 internal Monte Carlo replications and 200 bootstrap resamples. The narrowed and broadened baselines rescale the reconstructed posterior covariance by factors of $1/3$ and 2 , respectively.

final MIRA sample of $T = 16$ truths with $S = 160$ posterior draws per truth. These sources were selected according to computational availability, without reference to reconstruction quality. Each 256×256 truth and draw was flattened from $D = 65,536$ pixels and projected into a common 50-component PCA space, which retained 99.61 percent of the combined variance.

In the synthetic check, the matched Gaussian score was 0.6689, close to the finite-sample reference [10]. Reducing the covariance by a factor of three lowered the score to 0.6164, whereas doubling the covariance produced a score of 0.6678. The near equality of the matched and broadened scores shows that this check had limited sensitivity to a twofold increase in covariance.

The reconstructed posterior score is 0.0309 below the finite- S matched reference, a difference smaller than both uncertainty estimates in Table 4. At the available precision, MIRA provides no clear evidence of posterior miscalibration. This agreement is consistent with calibration but does not establish it.

Narrowing the reconstructed posterior lowers the score, while broadening it raises the score, producing the expected ordering of the point estimates. However, the substantial overlap of the one-standard-deviation uncertainty bars in Figure 8 shows that the diagnostic has limited sensitivity with only 16 truths. Each bootstrap resample also uses a newly generated internal region realisation, meaning that the bootstrap variance combines variation across truths with additional variation from region sampling.

All source runs use the same observation-noise realisation and aligned posterior random-number streams. The MIRA result is therefore conditional on this fixed simulation set. It does not cap-

ture the variation that would arise if the observations and posterior sample sets were regenerated independently.

5 Discussion

5.1 Reproducibility Verdict and Scope

The independent implementation reproduced the principal demonstrations of Adam et al. [6]. Unconditional samples exhibited galaxy-like morphologies, posterior samples recovered the dominant source structure, most image-plane residuals approached the assumed noise level, and the out-of-distribution experiment recovered the transition from likelihood-driven to prior-dominated inference. The core reproduction objectives were therefore met.

The multi-source residual analysis, PQMass, and MIRA extend this result by assessing statistical reliability beyond the original demonstrations. Within the 16 analysed sources, residual energy varied substantially and was strongly associated with lensed-image flux. Because the sources were selected according to computational availability, and because flux co-varies with apparent size and morphology, this relation should not be generalised to the wider PROBES population. PQMass found evidence against equality of the generated and selected PROBES distributions, whereas the MIRA score was compatible with its finite-sample matched reference at the available precision. These analyses identify source-dependent residual performance and a discrepancy between the generated and selected PROBES distributions, while preserving the conclusion that the principal demonstrations of Adam et al. were reproduced.

5.2 Prior Fidelity and Posterior Reliability

Visual plausibility does not establish distributional fidelity. Although the unconditional samples reproduced broad properties of the selected galaxies, they appeared generally smoother, and PQMass detected a discrepancy in both pixel and PCA representations [9]. This result applies to samples produced by the complete learned and numerical pipeline. It cannot by itself distinguish an inaccurate learned score from error introduced by the finite reverse process or from sensitivity to preprocessing and representation.

A mismatch in the generated distribution may be most consequential where the observation provides weak constraints. Strong lensing constrains different source-plane regions unevenly, while increasing the observation noise reduces the information supplied by the likelihood. The OOD experiment does not establish a causal connection between the PQMass discrepancy and reconstruction error. Instead, it illustrates how inaccuracies in the effective prior distribution could influence posterior reconstructions when the data are weak.

Image-plane residuals, PQMass, and MIRA address complementary questions [9, 10]. Residuals test consistency with the observation under the adopted forward and noise models. PQMass compares the generated and empirical source distributions, while MIRA assesses the location of each truth relative to its corresponding posterior sample set. A reconstruction can therefore agree with the observed image without establishing either distributional fidelity or posterior calibration.

In the present study, the MIRA score was compatible with its finite-sample matched reference at the available precision. The analysis used 16 distinct truths that shared one observation-noise realisation and aligned posterior random-number streams, so they should not be interpreted as 16 independent repetitions of the complete simulation procedure. The result is therefore conditional on the present simulation set. A stronger calibration assessment requires independently regenerated truth–observation pairs and coverage tests for scientifically interpretable quantities. Related studies of approximate inference have shown that apparently plausible posterior estimators can nevertheless be overconfident [29].

Several mechanisms may contribute to the observed discrepancies. Higher-flux sources may make small systematic errors more visible under a fixed noise variance, or they may contain more resolved

structure than the generated distribution represents accurately. The scalar likelihood approximation may also alter the relative influence of prior and observation. Average pooling changes the scale and correlation structure of the forward operator, so AA^T need not be proportional to the identity assumed by a scalar covariance treatment [6]. The PQMass result may additionally depend on reverse-process discretisation and on sensitivity to clipping, alignment, background structure, and compact contaminants. The present evidence does not separate these effects.

5.3 Scientific Value and Computational Cost

The scientific motivation for diffusion priors lies in their flexible representation of galaxy morphology and their potential to express non-Gaussian or multimodal source uncertainty [7, 6]. This capability is relevant to an ill-posed inverse problem in which a single posterior mean can conceal alternative plausible structures. Their potential benefit lies in representing uncertainty beyond a restrictive parametric source family or fixed regularisation scheme.

This flexibility incurs additional computational cost. Prior training can be amortised across multiple lens systems when the same trained model remains appropriate for the target population. Posterior sampling remains specific to each observation because every retained draw requires numerical integration of the reverse process. The scientific value of this additional computation should therefore be assessed through posterior calibration, empirical coverage, reconstruction accuracy, and downstream parameter recovery rather than through visual quality alone.

The present study did not include a matched parametric or regularised baseline. It therefore cannot determine whether diffusion-based inference provides more reliable uncertainty or improves downstream measurements relative to simpler alternatives. The results instead motivate a direct cost–performance comparison under common simulated observations and evaluation criteria. Additional computational expense is justified only if the broader posterior representation produces measurable gains in relevant source or lens quantities.

5.4 Future Work

Posterior validation should be strengthened using more independently generated truth–observation pairs. Each source should receive an independent observation-noise realisation and posterior sampling stream. MIRA could then be evaluated across multiple sources and noise levels, while simulation-based calibration could assess empirical coverage for interpretable quantities such as flux, size, ellipticity, orientation, and concentration [10, 24]. Increasing the number of independently regenerated pairs is more informative for this purpose than generating further posterior draws from the same observations.

The main numerical and modelling approximations should be isolated through controlled experiments. Prior samples should be generated across several reverse-process step counts to determine whether the PQMass discrepancy is sensitive to numerical integration [30]. Posterior reconstructions should likewise be compared across reverse-SDE discretisations. A pooling-aware likelihood covariance should be evaluated against the current scalar approximation and against likelihood treatments developed for noisy diffusion inverse problems [6, 31]. Source sets matched in flux but differing in morphology, and matched in morphology but differing in flux, would help identify which source properties are associated with larger residuals. These experiments would help separate learned-score mismatch, numerical sampling error, and error introduced by the likelihood approximation.

A matched cost–performance comparison is also needed. Representative alternatives include semi-linear pixelated inversion, Bayesian regularised source reconstruction, and adaptive grid-based modelling [5, 3, 4, 32]. These methods and the diffusion approach should be applied to the same independently generated truth–observation pairs. For methods that produce posterior samples, MIRA could compare conditional-distribution accuracy across approaches [10]. Deterministic reconstructions would require a probabilistic formulation before the same diagnostic could be applied.

Runtime, memory use, source error, image-plane residuals, empirical coverage, and downstream parameter recovery should be assessed separately, since MIRA does not measure computational cost

or point-estimate accuracy. Faster diffusion solvers and few-step generative methods suggest possible routes to reducing sampling cost [33, 34, 35, 36]. Most of these methods were not developed as direct replacements for the likelihood-conditioned reverse SDE used here and would require adaptation to posterior source reconstruction. Any accelerated implementation would then require direct comparison with the current sampler to establish that it preserves the relevant posterior target, including calibration, empirical coverage, and sample diversity.

Once these controlled tests are complete, the framework can be extended towards observational conditions. A realistic point-spread function would test sensitivity to instrumental blurring [37], while detector and reconstruction systematics could be assessed through controlled variations in noise, interpolation, and image processing [11]. Introducing lens-model uncertainty would extend the current fixed-lens analysis towards joint source and lens inference [38]. A held-out source population would test generalisation beyond the training distribution, while extending the framework to multi-band data would follow a direction proposed by Adam et al. [6]. Posterior diagnostics should also identify regimes in which the likelihood provides limited information and the reconstruction is determined primarily by the learned prior.

6 Conclusion

This project independently reproduced the principal results of the diffusion-based strong lensing source reconstruction method of Adam et al. [6]. Galaxy-like morphologies were recovered from the learned prior, posterior samples reconstructed the dominant source structure, and image-plane residuals approached the assumed noise level for most of the tested galaxies. The out-of-distribution experiment also recovered the reported transition from likelihood-driven reconstruction at low noise to prior-dominated behaviour as the observation became less informative. Together, these results establish the successful reproduction of the central method and its reported inference behaviour in the fixed-lens setting.

The validation extensions demonstrate why successful reproduction is an important starting point rather than the end of the investigation. Image-plane consistency varied between sources and was strongly associated with flux within the available sample. PQMass detected a difference between the generated prior and the selected PROBES images in both pixel and PCA representations, despite the visual plausibility of individual generated galaxies [9]. By contrast, the MIRA score of the reconstructed posterior was compatible with its finite-sample reference and provided no clear evidence of miscalibration at the available precision [10]. These findings do not overturn the reproduction result. Instead, they distinguish reproduction of the reported behaviour from the stronger requirements of prior fidelity, posterior calibration, and robustness across sources.

Diffusion models remain valuable because they can represent complex galaxy morphology and potentially non-Gaussian source uncertainty. Their scientific use, however, requires direct validation of prior fidelity, posterior calibration, and robustness under realistic observations. Validation is therefore the mechanism through which their expressive power can become trustworthy.

References

- [1] T. Treu, “Strong lensing by galaxies,” *Annual Review of Astronomy and Astrophysics*, vol. 48, no. 1, pp. 87–125, 2010.
- [2] T. E. Collett, “The population of galaxy–galaxy strong lenses in forthcoming optical imaging surveys,” *The Astrophysical Journal*, vol. 811, no. 1, p. 20, 2015.
- [3] S. H. Suyu, P. J. Marshall, M. P. Hobson, and R. D. Blandford, “A bayesian analysis of regularized source inversions in gravitational lensing,” *Monthly Notices of the Royal Astronomical Society*, vol. 371, pp. 983–998, sep 2006.
- [4] S. Vegetti and L. V. E. Koopmans, “Bayesian strong gravitational-lens modelling on adaptive grids: objective detection of mass substructure in galaxies,” *Monthly Notices of the Royal Astronomical Society*, vol. 392, pp. 945–963, jan 2009.
- [5] S. J. Warren and S. Dye, “Semi-linear gravitational lens inversion,” *The Astrophysical Journal*, vol. 590, no. 2, pp. 673–682, 2003.
- [6] A. Adam, A. Coogan, N. Malkin, R. Legin, L. Perreault-Levasseur, Y. Hezaveh, and Y. Bengio, “Posterior samples of source galaxies in strong gravitational lenses with score-based priors,” in *Machine Learning and the Physical Sciences Workshop at NeurIPS*, 2022.
- [7] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, “Score-based generative modeling through stochastic differential equations,” in *International Conference on Learning Representations*, 2021.
- [8] C. Stone, S. Courteau, N. Arora, M. Frosst, and T. H. Jarrett, “PROBES. I. A compendium of deep rotation curves and matched multiband photometry,” *The Astrophysical Journal Supplement Series*, vol. 262, p. 33, sep 2022.
- [9] P. Lemos, S. Sharief, N. Malkin, S. Salhi, C. Stone, L. Perreault-Levasseur, and Y. Hezaveh, “PQMass: Probabilistic assessment of the quality of generative models using probability mass estimation,” in *The Thirteenth International Conference on Learning Representations*, 2025.
- [10] S. Sharief, J. Zeghal, G. M. Barco, P. Lemos, Y. Hezaveh, and L. Perreault-Levasseur, “MIRA: A score for conditional distribution accuracy and model comparison,” 2026.
- [11] A. S. Tagore and C. R. Keeton, “Statistical and systematic uncertainties in pixel-based source reconstruction algorithms for gravitational lensing,” *Monthly Notices of the Royal Astronomical Society*, vol. 445, pp. 694–710, nov 2014.
- [12] W. R. Morningstar, L. Perreault Levasseur, Y. D. Hezaveh, R. Blandford, P. Marshall, P. Putzky, T. D. Rueter, R. Wechsler, and M. Welling, “Data-driven reconstruction of gravitationally lensed galaxies using recurrent inference machines,” *The Astrophysical Journal*, vol. 883, p. 14, Sept. 2019.
- [13] S. Mishra-Sharma and G. Yang, “Strong lensing source reconstruction using continuous neural fields,” in *Machine Learning for Astrophysics Workshop at the 39th International Conference on Machine Learning*, 2022.
- [14] B. Kavar, M. Elad, S. Ermon, and J. Song, “Denoising diffusion restoration models,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 23593–23606, Curran Associates, Inc., 2022.
- [15] K. Karchev, N. Anau Montel, A. Coogan, and C. Weniger, “Strong-lensing source reconstruction with denoising diffusion restoration models,” in *Machine Learning and the Physical Sciences Workshop at NeurIPS 2022*, 2022.

- [16] B. D. O. Anderson, “Reverse-time diffusion equation models,” *Stochastic Processes and their Applications*, vol. 12, pp. 313–326, May 1982.
- [17] A. Hyvärinen, “Estimation of non-normalized statistical models by score matching,” *Journal of Machine Learning Research*, vol. 6, no. 24, pp. 695–709, 2005.
- [18] P. Vincent, “A connection between score matching and denoising autoencoders,” *Neural Computation*, vol. 23, no. 7, pp. 1661–1674, 2011.
- [19] Y. Song and S. Ermon, “Generative modeling by estimating gradients of the data distribution,” in *Advances in Neural Information Processing Systems*, vol. 32, pp. 9184–9194, Curran Associates, Inc., 2019.
- [20] M. F. Naeem, S. J. Oh, Y. Uh, Y. Choi, and J. Yoo, “Reliable fidelity and diversity metrics for generative models,” in *Proceedings of the 37th International Conference on Machine Learning* (H. Daumé, III and A. Singh, eds.), vol. 119 of *Proceedings of Machine Learning Research*, pp. 7176–7185, PMLR, 2020.
- [21] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, “A kernel two-sample test,” *Journal of Machine Learning Research*, vol. 13, no. 25, pp. 723–773, 2012.
- [22] S. R. Cook, A. Gelman, and D. B. Rubin, “Validation of software for Bayesian models using posterior quantiles,” *Journal of Computational and Graphical Statistics*, vol. 15, no. 3, pp. 675–692, 2006.
- [23] P. Lemos, A. Coogan, Y. Hezaveh, and L. Perreault-Levasseur, “Sampling-based accuracy testing of posterior estimators for general inference,” in *Proceedings of the 40th International Conference on Machine Learning* (A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, eds.), vol. 202 of *Proceedings of Machine Learning Research*, pp. 19256–19273, PMLR, 2023.
- [24] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, “Validating Bayesian inference algorithms with simulation-based calibration,” 2018.
- [25] M. Modrák, A. H. Moon, S. Kim, P. Bürkner, N. Huurre, K. Faltejsková, A. Gelman, and A. Vehtari, “Simulation-based calibration checking for Bayesian computation: The choice of test quantities shapes sensitivity,” *Bayesian Analysis*, vol. 20, pp. 461–488, June 2025.
- [26] M. J. Smith, J. E. Geach, R. A. Jackson, N. E. Arora, C. Stone, and S. Courteau, “Realistic galaxy image simulation via score-based generative models,” *Monthly Notices of the Royal Astronomical Society*, vol. 511, pp. 1808–1818, apr 2022.
- [27] C. Stone, A. Adam, A. Coogan, M. J. Yantovski-Barth, A. Filipp, L. Setiawan, C. Core, R. Legin, C. Wilson, G. M. Barco, Y. Hezaveh, and L. Perreault-Levasseur, “Caustics: A python package for accelerated strong gravitational lensing simulations,” *Journal of Open Source Software*, vol. 9, no. 103, p. 7081, 2024.
- [28] R. Kormann, P. Schneider, and M. Bartelmann, “Isothermal elliptical gravitational lens models,” *Astronomy and Astrophysics*, vol. 284, pp. 285–299, apr 1994.
- [29] J. Hermans, A. Delaunoy, F. Rozet, A. Wehenkel, V. Begy, and G. Louppe, “A crisis in simulation-based inference? beware, your posterior approximations can be unfaithful,” *Transactions on Machine Learning Research*, 2022.
- [30] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 26565–26577, Curran Associates, Inc., 2022.

- [31] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. C. Ye, “Diffusion posterior sampling for general noisy inverse problems,” in *International Conference on Learning Representations*, 2023.
- [32] J. W. Nightingale and S. Dye, “Adaptive semi-linear inversion of strong gravitational lens imaging,” *Monthly Notices of the Royal Astronomical Society*, vol. 452, no. 3, pp. 2940–2959, 2015.
- [33] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “DPM-Solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 5775–5787, Curran Associates, Inc., 2022.
- [34] T. Salimans and J. Ho, “Progressive distillation for fast sampling of diffusion models,” in *International Conference on Learning Representations*, 2022.
- [35] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, “Consistency models,” in *Proceedings of the 40th International Conference on Machine Learning* (A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, eds.), vol. 202 of *Proceedings of Machine Learning Research*, pp. 32211–32252, PMLR, July 2023.
- [36] A. Mammadov, H. Chung, and J. C. Ye, “Amortized posterior sampling with diffusion prior distillation,” 2025.
- [37] A. Rogers and J. D. Fiege, “Strong gravitational lens modeling with spatially variant point-spread functions,” *The Astrophysical Journal*, vol. 743, no. 1, p. 68, 2011.
- [38] K. Karchev, A. Coogan, and C. Weniger, “Strong-lensing source reconstruction with variationally optimized gaussian processes,” *Monthly Notices of the Royal Astronomical Society*, vol. 512, no. 1, pp. 661–685, 2022.

A Auto-generation Usage

OpenAI’s ChatGPT and Codex, together with Anthropic’s Claude Code, were used as supporting tools during the development of this project. Their use included discussing the structure of the report, suggesting revisions for clarity and concision, checking consistency between sections, and assisting with the presentation of mathematical notation. The tools were also used to review Python and L^AT_EX code, suggest debugging steps, inspect consistency between the implementation and its description, and help organise numerical outputs into figures and tables.

Outputs from these tools were treated as suggestions rather than authoritative results. The author ran all training, posterior-sampling, and validation experiments and checked the numerical values reported in the text against saved outputs. Suggested code changes were inspected and executed before use. Bibliographic suggestions were included only after checking the original papers and their metadata. The interpretation of the results, selection of evidence, and final conclusions remain the responsibility of the author.

A generated image was used as the source in the qualitative out-of-distribution experiment. It was created using OpenAI’s image-generation tool in ChatGPT with the prompt:

“A galaxy in the shape of the number 7 on a dark background.”

This matches the prompt reported by Adam et al. The image was used only as an input to the prompt-matched qualitative noise experiment. No generated image was included in the prior-training data or used as a real astronomical observation.